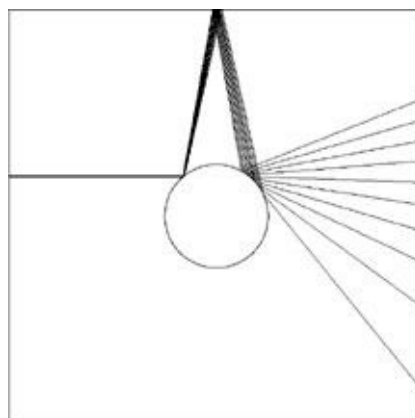


us its own limits rather clearly.

There is (as usual) an element of the unexpected in this story. Poincaré initially responded to a competition organized by the mathematician Gösta Mittag-Leffler to celebrate the sixtieth birthday of King Oscar of Sweden. Poincaré's memoir, which was about the stability of the solar system, won the prize that was then the highest scientific honor (as these were the happy days before the Nobel Prize). A problem arose, however, when a mathematical editor checking the memoir before publication realized that there was a calculation error, and that, after consideration, it led to the opposite conclusion—unpredictability, or, more technically, nonintegrability. The memoir was discreetly pulled and reissued about a year later.

Poincaré's reasoning was simple: as you project into the future you may need an increasing amount of precision about the dynamics of the process that you are modeling, since your error rate grows very rapidly. The problem is that near precision is not possible since the degradation of your forecast compounds abruptly—you would eventually need to figure out the past with infinite precision. Poincaré showed this in a very simple case, famously known as the “three body problem.” If you have only two planets in a solar-style system, with nothing else affecting their course, then you may be able to indefinitely predict the behavior of these planets, no sweat. But add a third body, say a comet, ever so small, between the planets. Initially the third body will cause no drift, no impact; later, with time, its effects on the two other bodies may become explosive. Small differences in where this tiny body is located will eventually dictate the future of the behemoth planets.

FIGURE 2: PRECISION AND FORECASTING



One of the readers of a draft of this book, David Cowan, gracefully drew this picture of scattering, which shows how, at the second bounce, variations in the initial conditions can lead to extremely divergent results. As the initial imprecision in the angle is multiplied, every additional bounce will be further magnified. This causes a severe multiplicative effect where the error grows out disproportionately.